CHAPTER EXERCISE: Answers for this chapter start on page 301.

A calculator should NOT be used on the following questions.

1

Which of the following represents the square of the sum of x and y, decreased by the product of x and y ?

A) $x^2 + y^2 - xy$
B) $x^2y^2 - xy$
C) $(x+y)^2 - (x+y)$
D) $(x+y)^2 - xy$

2

On a 100 cm ruler, lines are drawn at 10, X, and 98 cm. The distance between the lines at X and 98 cm is three times the distance between the lines at X and 10 cm. What is the value of X ?

3

If 5 is added to the square root of x, the result is 9. What is the value of $x + 2$?

4

A grocery store sells tomatoes in boxes of 4 or 10. If Melanie buys x boxes of 4 and y boxes of 10, where $x \geq 1$ and $y \geq 1$, for a total of 60 tomatoes, what is one possible value of x ?

5

A rectangular monitor has a length of x inches and a width that is one-third of its length. If the perimeter of the monitor is 48 inches, what is the value of x ?

6

Susie buys 2 pieces of salmon, each weighing x pounds, and 1 piece of trout, weighing y pounds, where x and y are integers. The salmon cost \$3.50 per pound and the trout cost \$5 per pound. If the total cost of the fish was \$77, which of the following could be the value of y ?

A) 4
B) 5
C) 6
D) 7

7

A 20% nickel alloy was made by combining 2 grams of a 35% nickel alloy with 6 grams of an x% nickel alloy. What is the value of x ?

A calculator is allowed on the following questions.

8

If $8 + 5x$ is twice $x - 5$, what is the value of x ?

A) -6

B) -3

C) $-\frac{7}{3}$

D) -2

9

If 75% of 68 is the same as 85% of n, what is the value of n ?

10

The Pirates won exactly 4 of their first 15 games. They then played N remaining games and won all of them. If they won exactly half of all the games they played, what is the value of N ?

11

Alice and Julie start with the same number of pens. After Alice gives 16 of her pens to Julie, Julie then has two times as many pens as Alice does. How many pens did Alice have at the start?

12

At a Hong Kong learning center, $\frac{1}{4}$ of the students take debate, $\frac{1}{6}$ of the students take writing, and $\frac{1}{8}$ of the students take science. The rest take math. If 33 students take math, what is the total number of students at the learning center?

A) 60

B) 66

C) 72

D) 78

13

Ian has 20 football cards, and Jason has 44 baseball cards. They agree to trade such that Jason gives Ian 2 baseball cards for every card Ian gives to Jason. After how many such trades will Ian and Jason each have an equal number of cards?

A) 9

B) 10

C) 11

D) 12

14

If 3 is subtracted from 3 times the number x, the result is 21. What is the result when 8 is added to half of x ?

A) 1

B) 5

C) 8

D) 12

15

At a store, the price of a tie is k dollars less than three times the price of a shirt. If a shirt costs \$40 and a tie costs \$30, what is the value of k ?

16

A wooden board in the shape of a rectangle has a length that is twice its width. If the area of the board is 128 square feet, what is the length, in feet, of the board?

17

Alex, Bob, and Carl all collect seashells. Bob has half as many seashells as Carl. Alex has three times as many seashells as Bob. If Alex and Bob together have 60 seashells, how many seashells does Carl have?

A) 15
B) 20
C) 30
D) 40

18

Mark and Kevin own $\frac{1}{4}$ and $\frac{1}{3}$ of the books on a shelf, respectively. Lori owns the rest of the books. If Kevin owns 9 more books than Mark, how many books does Lori own?

19

A bakery gave out coupons to celebrate its grand opening. Each coupon was worth either \$1, \$3, or \$5. Twice as many \$1 coupons were given out as \$3 coupons, and 3 times as many \$3 coupons were given out as \$5 coupons. The total value of all the coupons given out was \$360. How many \$3 coupons were given out?

A) 40
B) 45
C) 48
D) 54

20

A water tank is connected to two pipes, Pipe A and Pipe B. It takes 4 hours to fill the tank when only Pipe A is in use, and it takes 6 hours to fill the tank when only Pipe B is in use. If it takes m minutes to fill the tank when both Pipe A and Pipe B are in use, what is the value of m ?

21

Yoona runs at a steady rate of 1 yard per second. Jessica runs 4 times as fast. If Jessica gives Yoona a head start of 30 yards in a race, how many yards must Jessica run to catch up to Yoona?

22

Nicky owns a house that has a patio in the shape of a square. She decides to renovate the patio by increasing its length by 4 feet and decreasing its width by 5 feet. If the area of the renovated patio is 90 square feet, what was the original area of the patio, in square feet?

23

Terry is hired to pave a parking lot and finishes $\frac{1}{3}$ of the parking lot before Andy is hired to work alongside him. They each work at a constant rate, but Terry works twice as fast as Andy does. The equation $9\left(\frac{1}{x}+\frac{1}{2x}\right)=\frac{2}{3}$ can be used to find the total number of days x it would have taken Terry to pave the entire parking lot by himself. Which of the following is the best interpretation of the number 9 in the equation?

A) The number of days it would have taken Terry and Andy to pave the entire parking lot if they had worked together from the start.

B) The number of days it will take Terry and Andy to pave the remainder of the parking lot working together.

C) The number of days it would take Andy to pave the remainder of the parking lot if he were working alone.

D) The number of days it would take Terry to pave the remainder of the parking lot if he were working alone.

Chapter 12: Word Problems

CHAPTER EXERCISE:

1. $\boxed{D}$ The square of the sum of x and y is $(x+y)^2$. The product of x and y is xy. The question asks for the difference: $(x+y)^2 - xy$

2. $\boxed{32}$

$$\begin{aligned} 98 - X &= 3(X - 10) \\ 98 - X &= 3X - 30 \\ -4X &= -128 \\ X &= 32 \end{aligned}$$

3. $\boxed{18}$

$$\begin{aligned} \sqrt{x} + 5 &= 9 \\ \sqrt{x} &= 4 \\ x &= 16 \end{aligned}$$

$x + 2 = 18$

4. $\boxed{5 \text{ or } 10}$ Based on the information, we can form the equation $4x + 10y = 60$. Now it's just a matter of guess and check. Since x and y are integers, it won't be long before we find something that works. For example, $x = 5, y = 4$ is one possible solution.

5. $\boxed{18}$ The width of the monitor is $\frac{1}{3}x$. Since the perimeter of a rectangle is twice the length plus twice the width,

$$\begin{aligned} 2x + 2\left(\frac{1}{3}x\right) &= 48 \\ 2x + \frac{2}{3}x &= 48 \\ \frac{8}{3}x &= 48 \\ x = 48 \cdot \frac{3}{8} &= 18 \end{aligned}$$

6. $\boxed{D}$ Susie bought $2x$ pounds of salmon and y pounds of trout. The total cost is then

$$(3.50)(2x) + 5y = 77$$

$$7x + 5y = 77$$

Since x and y must be integers, we can plug each answer choice into the equation above to see if we get an integer value for x. When $y = 4$, for example, $x \approx 8.14$, which is not an integer. The answer turns out to be 7. When $y = 7, x = 6$.

7. $\boxed{15}$ The 35% nickel alloy contains $2(0.35) = 0.7$ grams of nickel. The x% nickel alloy contains $\frac{x}{100} \cdot 6 = 0.06x$ grams of nickel. When combined, these alloys formed $2 + 6 = 8$ grams of a 20% nickel alloy, which must contain $8(0.20) = 1.6$ grams of nickel. Setting up an equation, we get

$$\begin{aligned} 0.7 + 0.06x &= 1.6 \\ 0.06x &= 0.9 \\ 6x &= 90 \\ x &= 15 \end{aligned}$$

8. $\boxed{A}$

$$\begin{aligned} 8 + 5x &= 2(x - 5) \\ 8 + 5x &= 2x - 10 \\ 3x &= -18 \\ x &= -6 \end{aligned}$$

9. [60] Converting 75% and 85% to fractions in the equation below,

$$\begin{aligned}\frac{3}{4}(68) &= \frac{17}{20}n \\ 51 &= \frac{17}{20}n \\ 51\left(\frac{20}{17}\right) &= n \\ 60 &= n\end{aligned}$$

10. [7]

$$\frac{4+N}{15+N} = \frac{1}{2}$$

Cross multiplying,

$$\begin{aligned}2(4+N) &= 15+N \\ 8+2N &= 15+N \\ N &= 7\end{aligned}$$

11. [48] They start with the same number x. Once Alice gives 16 to Julie, Alice is left with $x-16$ and Julie then has $x+16$.

$$\begin{aligned}x+16 &= 2(x-16) \\ x+16 &= 2x-32 \\ -x &= -48 \\ x &= 48\end{aligned}$$

12. [C] The fraction of students who take math is $1-\frac{1}{4}-\frac{1}{6}-\frac{1}{8}=\frac{11}{24}$. Let x be the total number of students.

$$\begin{aligned}\frac{11}{24}x &= 33 \\ x &= 72\end{aligned}$$

13. [D] Let x be the number of trades. Each trade, Ian has a net gain of 1 card while Jason has a net loss of 1 card.

$$\begin{aligned}20+x &= 44-x \\ 2x &= 24 \\ x &= 12\end{aligned}$$

14. [D] Making an equation to figure out x,

$$\begin{aligned}3x-3 &= 21 \\ 3x &= 24 \\ x &= 8\end{aligned}$$

$$8+\frac{1}{2}(8) = 12$$

15. [90] Three times the price of a shirt is 120. Since a tie, which costs 30, is k less than that, k must be $120-30=90$. As an equation,

$$\begin{aligned}\text{tie} &= 3(\text{shirt})-k \\ 30 &= 3(40)-k \\ 30 &= 120-k \\ k &= 90\end{aligned}$$

16. [16] Let the width of the board be w. Then its length is $2w$. Since the area of a rectangle is its length times its width,

$$\begin{aligned}(2w)(w) &= 128 \\ 2w^2 &= 128 \\ w^2 &= 64 \\ w &= \sqrt{64} = 8\end{aligned}$$

Finally, the length is $2w = 2(8) = 16$.

17. [C] Let x be the number of seashells that Carl has. Bob then has $\frac{1}{2}x$ seashells and Alex has $\frac{3}{2}x$ seashells. Since Alex and Bob together have 60 seashells,

$$\begin{aligned}\frac{1}{2}x+\frac{3}{2}x &= 60 \\ 2x &= 60 \\ x &= 30\end{aligned}$$

Carl has 30 seashells.

18. $\boxed{45}$ Let x be the total number of books. Mark then has $\frac{1}{4}x$ books and Kevin has $\frac{1}{3}x$ books. Kevin owns 9 more than Mark, so

$$\frac{1}{3}x - \frac{1}{4}x = 9$$

Multiplying both sides by 12,

$$4x - 3x = 108$$
$$x = 108$$

The total number of books is 108. Mark owns $\frac{1}{4} \times 108 = 27$ books and Kevin owns $\frac{1}{3} \times 108 = 36$ books. Lori must then own $108 - 27 - 36 = 45$ books.

19. $\boxed{D}$ Let the number of \$5 coupons given out be x. Then the number of \$3 coupons given out is $3x$, and the number of \$1 coupons given out is $2(3x) = 6x$.

$$5(x) + 3(3x) + 1(6x) = 360$$
$$5x + 9x + 6x = 360$$
$$20x = 360$$
$$x = 18$$

The number of \$3 coupons given out is then $3x = 3(18) = 54$.

20. $\boxed{144}$ Pipe A by itself can fill $\frac{1}{4}$ of the tank each hour, and Pipe B by itself can fill $\frac{1}{6}$ of the tank each hour. When used together, they can fill $\frac{1}{4} + \frac{1}{6} = \frac{5}{12}$ of the tank each hour. Now we can use the formula $W = rt$, where W is 1 tank and $r = \frac{5}{12}$ of the tank per hour, to find t.

$$1 = \frac{5}{12}t$$
$$\frac{12}{5} = t$$

Therefore, it takes $\frac{12}{5}$ hours for both pipes to fill the tank. Converting this result to minutes, we get $m = \frac{12}{5} \times 60 = 144$.

21. $\boxed{40}$ Jessica runs at a rate of 4 yards per second. Let t be the time it takes for Jessica to overtake Yoona. We can make an equation with the left side being Yoona's distance and the right side being Jessica's distance.

$$30 + t = 4t$$
$$30 = 3t$$
$$10 = t$$

It takes 10 seconds for Jessica to catch up to Yoona. In that time, Jessica runs $4(10) = 40$ yards.

22. $\boxed{121}$ Let the side length of the original patio be s. Then the renovated patio has a length of $s + 4$ and a width of $s - 5$. Setting up an equation for the area of the renovated patio, we get

$$(s + 4)(s - 5) = 90$$
$$s^2 - s - 20 = 90$$
$$s^2 - s - 110 = 0$$
$$(s - 11)(s + 10) = 0$$

Since the side length s must be a positive number, $s = 11$, which means the original area of the patio was $s^2 = (11)^2 = 121$ sq ft.

23. $\boxed{B}$ This problem follows the format of a standard $W = rt$ word problem, where W is the amount of work done, r is the overall rate at which work is being done, and t is the time spent. If Terry can finish the parking lot in x days by himself, then Andy can finish it in $2x$ days by himself. This means that Terry paves $\frac{1}{x}$ of the parking lot each day and Andy paves $\frac{1}{2x}$ of the parking lot each day. Working together, they pave $\frac{1}{x} + \frac{1}{2x}$ of the parking lot each day. This is the overall rate r. The number $\frac{2}{3}$ is the remaining fraction of the parking lot that has yet to be paved. This is W. So the number 9 must be t. And given what we have for W and r, t must be the number of days it will take Terry and Andy to pave the remainder of the parking lot working together.